Numpy program

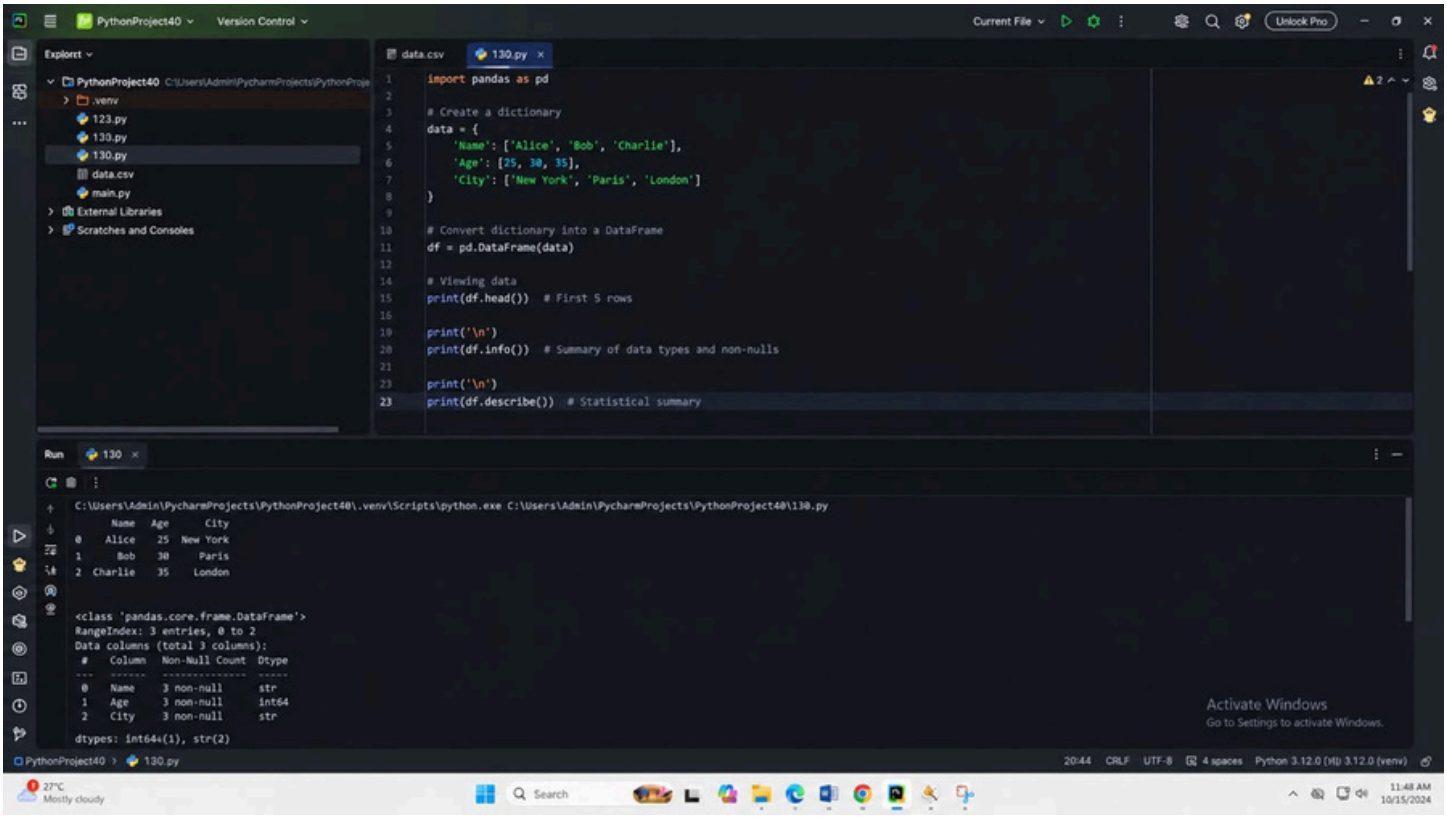
Aim

TowriteaPython programusing the NumPy library to create and display 1D and 2D arrays,

zerosarray,ones array,identity matrix, and range of values.

Algorithm

1. **Starttheprogram.**
2. **Import the NumPylibrary using import numpy as np.**
3. **Create and displaya1Darray containing elements from 1 to 5.**
4. **Create and displaya2Darray with 2 rows and 3 columns.**
5. **Create and displaya\(3\times 3\) array filled with zeros.**
6. **Create and displaya\(2\times 4\) array filled with ones.**
7. **Create and displaya\(3\times 3\) identity matrix.**
8. **Generate and displaynumbers from 0 to 10 with a step size of 2 using np.arange().**
9. **Stop the program**



Result

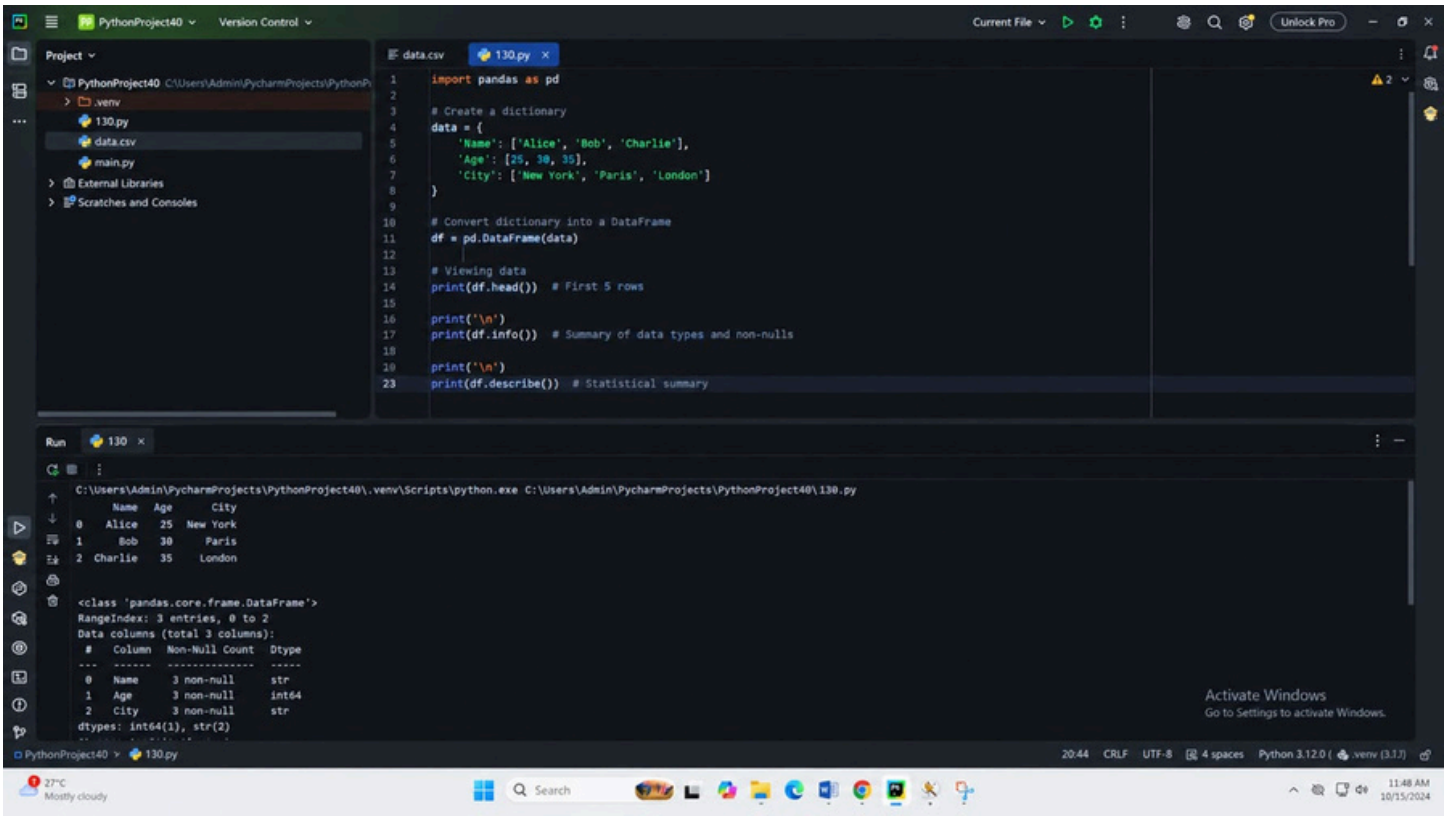
Thus, the Python program using NumPy was executed successfully, and the 1D array, 2D array, zeros array, ones array, identity matrix, and range of values were displayed.

Aim

To write aPython program using NumPy to perform array addition, multiplication, scalar multiplication, square root, mean, and dot product operations.

Algorithm

1. **Start the program.**
2. **Import the NumPylibrary using import numpy as np.**
3. **Create two arraysa=[1,2, 3] and b = [4, 5, 6].**
4. **Display both arrays.**
5. **Perform and displayarray addition using a + b.**
6. **Perform and displayelement-wise multiplication using a * b.**
7. **Perform scalar multiplication using a * 10.**
8. **Calculate and displaythe square root of array a using np.sqrt(a).**
9. **Calculate and displaythe mean of array a using np.mean(a).**
10. **Calculate and displaythe dot product of arrays a and b using np.dot(a, b).**
11. **Stop the program.**



Result

Thus, the Python program using NumPy was executed successfully, and array addition, element-wise multiplication, scalar multiplication, square root, mean, and dot product operations were performed

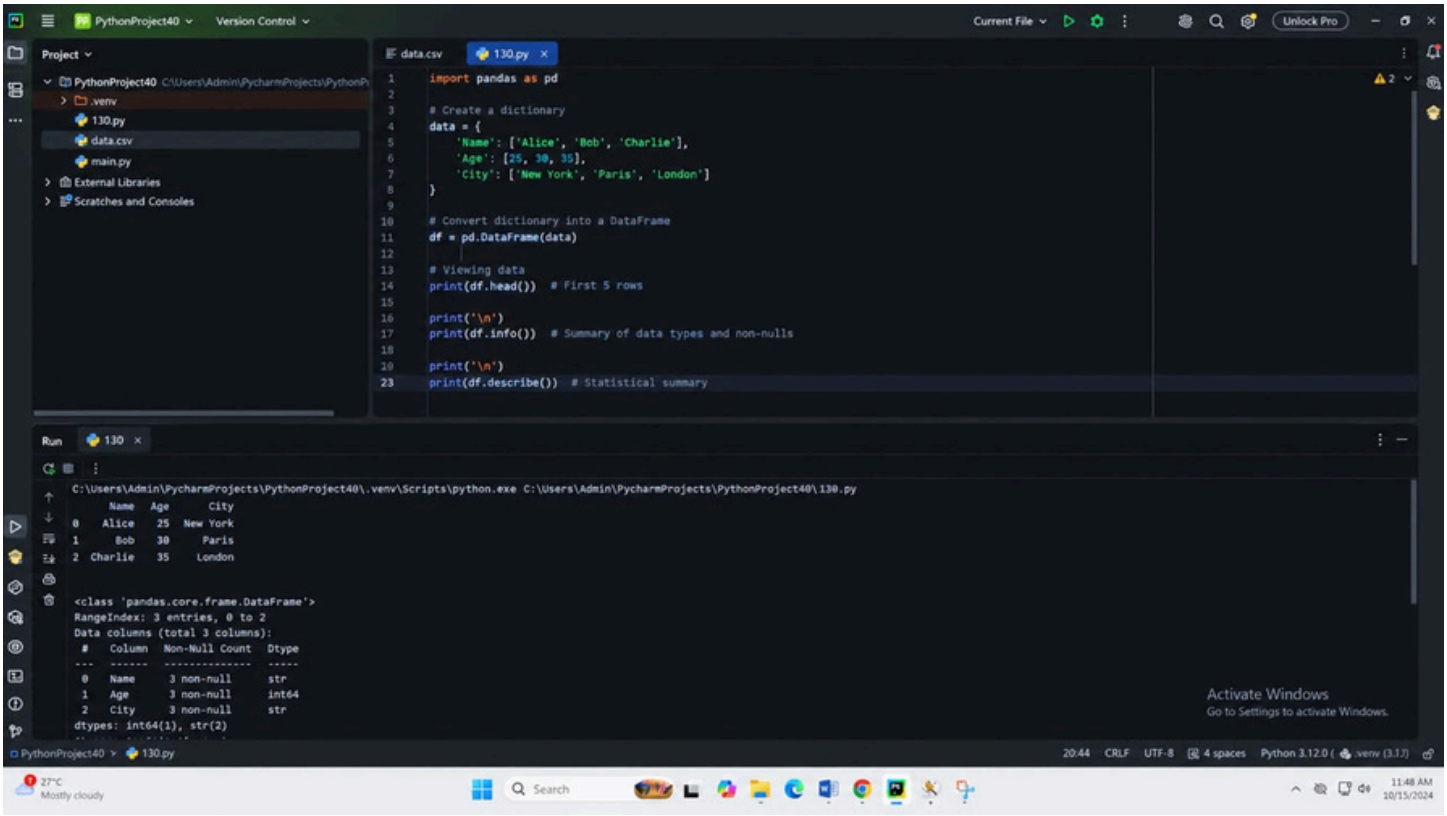
Aim

To write aPython program using NumPy to perform matrix indexing, row selection, column selection,and sub-matrix slicing.

Algorithm

1. **Start the program.**
2. **Import the NumPy library using import numpy as np.**
3. **Create a \(3 \times 3\) matrix with the elements 10 to 90.**

4. Display the matrix.
5. Access and display the element at index [1, 2].
6. Select and display the first row using matrix[0, :].
7. Select and display the second column using matrix[:, 1].
8. Extract and display the top-left \(2 \times 2\) sub-matrix using matrix[0:2, 0:2].
9. Stop the program.



Result

Thus, the Python program using NumPy was executed successfully, and matrix indexing, row selection, column selection, and sub-matrix slicing were performed.